

EDUGENIE-GOOGLE GEMINI POWERED LEARNING ASSISTANT

IDEATION DOCUMENT

Project Title: EduGenie – Google Gemini Powered Learning Assistant

Project Type: Generative AI / Educational Assistant

Repository: deepukutty9812-ai/EduGenie-AI

Development Platform: Python, FastAPI, Jinja2, Google Gemini

Document: Project Ideation

1. Introduction

EduGenie is a lightweight, AI-powered educational assistant designed to support students in their learning process using Generative Artificial Intelligence.

The system provides multiple educational services through a single web application. Instead of using separate tools for answering questions, explaining concepts, generating quizzes, summarizing study material, and planning learning activities, EduGenie combines these functions into one simple interface.

The current implementation uses **FastAPI and Jinja2** for the web application and integrates the **Google Gemini generative AI model** for most AI-powered functions. The project also supports an optional **LaMini-Flan-T5-783M local model** for concept explanation, with Gemini available as a fallback.

The main functions implemented in the repository are:

1. Question and Answer
2. Concept Explanation
3. Three-Question MCQ Quiz Generation
4. Passage Summarization
5. Personalized Learning Path Recommendation

These functions are exposed through dedicated FastAPI endpoints.

2. Problem Identification

Students frequently face difficulties while learning because educational assistance is distributed across different resources.

Common problems include:

- Difficulty understanding complex academic concepts.
- Lack of immediate answers to study-related questions.
- Difficulty creating practice questions from study material.
- Spending considerable time reading and revising lengthy passages.
- Lack of structured guidance when learning a new topic.
- Need for explanations appropriate to different learner levels.
- Switching between multiple applications for different learning activities.

Traditional learning resources may provide information, but they do not always provide an interactive and personalized learning experience.

Therefore, there is a need for an intelligent educational assistant capable of responding to different learning requirements through a single interface.

3. Proposed Idea

The proposed idea is to develop **EduGenie**, a Generative AI-based educational assistant that allows students to perform multiple learning activities from one web application.

The learner provides a question, topic, or passage and selects the required operation.

The system then processes the request and generates an appropriate educational response.

The overall concept is:

Student Input → Task Selection → FastAPI Backend → AI Processing → Educational Response

The application uses Gemini for question answering, quiz generation, summarization, and learning recommendations. Concept explanation can use the optional local LaMini-Flan-T5-783M model or Gemini fallback.

4. Aim of the Project

The main aim of EduGenie is to develop a simple and accessible AI-powered learning assistant that helps students understand, practise, revise, and plan their learning using Generative AI.

The project aims to demonstrate the practical application of Large Language Model technology in education.

5. Objectives

The major objectives of the project are:

1. To develop an AI-powered educational assistant using Generative AI.
 2. To provide students with quick answers to academic questions.
 3. To simplify difficult concepts according to the learner's level.
 4. To automatically generate multiple-choice questions for practice.
 5. To summarize lengthy educational content into concise material.
 6. To provide structured learning recommendations for selected topics.
 7. To integrate different educational functions into one web application.
 8. To provide a lightweight application that can be executed locally.
 9. To design the backend in a modular manner so that additional educational features can be added later.
 10. To demonstrate the integration of an AI model with a modern Python web framework.
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6. Target Users

EduGenie is primarily intended for:

6.1 Students

Students can use EduGenie to clear doubts, understand concepts, practise questions, and summarize study material.

6.2 Self-Learners

People learning independently can use the learning-path feature to obtain structured recommendations.

6.3 Beginners

Beginners can request simplified explanations of unfamiliar concepts.

6.4 Academic Demonstration

The project can also be used as an academic demonstration of Generative AI, REST APIs, prompt engineering, and web application development.

7. Core Features

7.1 Question and Answer

The Q&A feature allows the user to enter a question and receive an AI-generated response.

The corresponding backend endpoint is:

POST /qa

Example:

Input:

"Which is the largest ocean?"

Output:

An AI-generated answer explaining the relevant information.

This feature is intended for quick doubt clarification.

7.2 Concept Explanation

The explanation feature allows the learner to provide a topic and learning level.

The corresponding endpoint is:

POST /explain

The supported learner levels include:

- Beginner
- Intermediate
- Advanced

Example:

Topic: Pythagoras theorem

Level: Beginner

The system generates an explanation appropriate to the selected level.

The project documentation specifies the optional **LaMini-Flan-T5-783M** model for this functionality. In the current implementation, the local model is optional and Gemini can be used as a fallback.

7.3 Quiz Generation

EduGenie can generate a quiz from a supplied passage or topic.

The endpoint is:

POST /quiz

The current project design generates:

- 3 multiple-choice questions
- 4 options for each question
- One correct answer for each question

The generated quiz is requested in structured JSON format, making it easier for the application to process and display the questions.

This feature supports active recall and self-assessment.

7.4 Text Summarization

The summarization feature converts a longer passage into a concise form.

The endpoint is:

POST /summarize

The objective is to retain the important information while reducing unnecessary length.

This can be useful for:

- Revision
 - Quick reading
 - Exam preparation
 - Understanding lengthy study material
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7.5 Personalized Learning Path

The learning-path feature generates recommendations for learning a particular topic.

The endpoint is:

POST /learn/recommendations

The user can provide:

- Topic
- Learning level

Example:

Topic: SQL

Level: Beginner

The system generates a structured learning path that can guide the learner from fundamental concepts toward more advanced topics.

8. Generative AI Component

Generative AI is the central technology of EduGenie.

Instead of storing predefined answers for every possible question, the system sends suitable prompts to the configured AI model and generates responses dynamically.

The Gemini model is used for:

- Question answering
- Quiz generation
- Summarization
- Learning recommendations

The current repository uses the Google GenAI Python SDK and configures gemini-2.5-flash as the default Gemini model. The README notes that the model is configurable and supports structured output.

For concept explanation, the project additionally supports the optional local LaMini-Flan-T5-783M model.

9. Proposed System Workflow

The basic workflow of EduGenie is:

Step 1 – User Access

The learner opens the EduGenie web application.

Step 2 – Task Selection

The learner selects one of the available operations:

- Q&A

- Explain
- Quiz
- Summarize
- Recommend Learning Path

Step 3 – Input

The learner enters the required question, topic, or text.

Step 4 – API Request

The browser sends the request to the appropriate FastAPI endpoint.

Step 5 – AI Processing

The corresponding module processes the input and communicates with Gemini or the optional local explanation model.

Step 6 – Response Generation

The AI generates the requested educational content.

Step 7 – Result Display

The generated result is returned to the web interface and displayed to the learner.

10. High-Level Architecture Idea

The repository describes the architecture as:

Browser

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FastAPI + Jinja2

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Learning API Endpoints

- /qa → Gemini
- /explain → Optional LaMini local model / Gemini fallback
- /quiz → Gemini structured JSON
- /summarize → Gemini
- /learn/recommendations → Gemini



Generated Educational Response

This architecture keeps the user interface, API routing, and AI processing logically separated.

11. Technology Selection

The project uses the following major technologies:

Technology	Purpose
Python	Main programming language
FastAPI	Backend web framework and REST API
Uvicorn	ASGI application server
Jinja2	HTML templating
HTML	Frontend structure
CSS	Frontend styling
Google GenAI SDK	Gemini API integration
Gemini	Main Generative AI model
LaMini-Flan-T5-783M	Optional local explanation model
Pydantic	Request and response validation
Pytest	Automated testing

The repository requires **Python 3.10+** and provides a local virtual-environment setup using requirements.txt.

12. Innovation of the Idea

The main innovation of EduGenie is not simply providing a chatbot.

Instead, the system organizes Generative AI into several specific educational functions.

The same AI-based application can act as:

- A doubt-solving assistant
- A concept explainer

- A quiz generator
- A summarization assistant
- A learning-path advisor

This task-oriented approach makes Generative AI more useful for everyday student learning.

13. Example Use Cases

Use Case 1 – Academic Doubt

A student wants to know the answer to an academic question.

The student selects **Q&A**, enters the question, and receives an AI-generated response.

Use Case 2 – Difficult Concept

A beginner wants to understand the Pythagoras theorem.

The student selects **Explain**, enters the topic and beginner level, and receives a simplified explanation.

Use Case 3 – Self Assessment

A student has completed reading a topic and wants to test their understanding.

The student selects **Quiz**, enters the relevant passage, and receives three MCQs.

Use Case 4 – Revision

A student has a lengthy paragraph and wants to revise it quickly.

The student selects **Summarize** and receives a concise version.

Use Case 5 – Learning a New Technology

A beginner wants to learn SQL.

The student selects **Recommend Path**, enters SQL and beginner level, and receives a structured learning path.

14. Expected Benefits

The expected benefits of EduGenie are:

- Faster access to educational assistance.
- Simplified understanding of difficult concepts.
- Easy generation of practice questions.

- Faster revision through summarization.
 - Structured guidance for learning new subjects.
 - One interface for multiple learning activities.
 - Demonstration of practical Generative AI integration.
 - Modular architecture suitable for future expansion.
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15. Project Scope

The current project focuses on five major educational functions:

1. Question answering
2. Concept explanation
3. Quiz generation
4. Text summarization
5. Learning-path recommendations

The application is intentionally lightweight and suitable for local execution.

The repository provides endpoints, API documentation through FastAPI, a health-check endpoint, and automated tests that mock Gemini so the tests do not require a paid API call.

16. Current Limitations

The current implementation is intentionally lightweight.

The following features are not part of the current core implementation:

- User authentication
- Persistent user accounts
- Long-term learner progress storage
- Production-grade rate limiting
- Persistent learning analytics
- Full production deployment infrastructure

The repository identifies authentication, rate limiting, logging, secret management, HTTPS, request-size limits, and persistent user/progress storage as production considerations.

17. Future Enhancement Ideas

Possible future enhancements include:

1. User registration and authentication.
2. Student progress tracking.
3. Learning analytics dashboard.
4. Adaptive learning recommendations.
5. Voice-based interaction.
6. Multilingual educational assistance.
7. Mobile application.
8. Gamification and achievement tracking.
9. Image-based question solving.
10. PDF/document-based learning.
11. Persistent quiz history.
12. Teacher and student dashboards.
13. Integration with Learning Management Systems.
14. Personalized study schedules.
15. Advanced assessment and performance analysis.

These enhancements can extend EduGenie from a lightweight learning assistant into a more comprehensive intelligent learning platform.

18. Expected Project Outcome

The expected outcome of the project is a functional web-based Generative AI educational assistant.

The completed system should allow learners to:

- Ask questions.
- Understand concepts.
- Generate quizzes.
- Summarize text.
- Obtain learning recommendations.

The project also demonstrates important technical concepts including:

- Generative AI
- Prompt engineering
- REST API development
- FastAPI
- Web application development
- AI model integration
- Structured AI output
- Request/response validation
- Automated testing

19. Conclusion

EduGenie is conceived as a learner-focused Generative AI educational assistant that combines multiple common study activities into a single lightweight application.

The project addresses the need for quick, interactive, and adaptable learning support by allowing users to ask questions, obtain simplified explanations, generate quizzes, summarize content, and receive structured learning recommendations.

The current repository implements this concept using FastAPI, Jinja2, Google Gemini, and optional local LaMini-Flan-T5 support. Its modular architecture provides a practical foundation for future additions such as authentication, progress tracking, multilingual learning, voice interaction, and advanced personalized learning.

Thus, EduGenie demonstrates how Generative AI can be applied to create a practical educational application that supports understanding, practice, revision, and guided learning.